Distributed Microservices Based E-Commerce Platform

ABSTRACT

The objective of this project is to build a scalable and modular e-commerce platform using distributed microservices. It provides key features such as product catalog, shopping cart, order processing, and secure authentication. The goal is to showcase Java Full Stack development skills with modern frameworks while solving real-world online commerce challenges.

EXISTING SYSTEM

Traditional e-commerce applications are often built as monolithic systems, where all modules such as product management, cart, orders, and authentication are tightly coupled. This design leads to several limitations:

- Difficult to scale individual modules when user demand increases.
- Maintenance becomes complex as the application grows.
- Deployment cycles are slower because any change requires rebuilding the entire system.
- Limited flexibility in integrating new technologies or services.

Such systems struggle to meet the growing demand for high availability, modularity, and faster development cycles in modern IT industries.

PROPOSED SYSTEM

The proposed system adopts a distributed microservices architecture, where each module (Product Service, Cart Service, Order Service, Authentication Service) is developed and deployed independently. These services communicate through RESTful APIs, ensuring flexibility, scalability, and maintainability.

Key features of the proposed system:

- **User Authentication & Role Management:** Secure login and registration for customers and administrators.
- **Product Catalog:** CRUD operations for products with category and search functionality.
- **Shopping Cart:** Add/remove items, update quantities, and calculate totals.
- **Order Management:** Place orders, track status, and simulate payment workflows.
- **Admin Dashboard:** Manage products, view orders, and monitor inventory.
- **Responsive Frontend:** Built with React.js and Bootstrap for a modern user experience.

This architecture allows independent scaling of services, faster deployment, and easier integration with third-party modules such as payment gateways or notification systems.

USERS

The system is designed for two primary user groups:

1. CUSTOMERS

- Register/login to the platform.
- Browse products, add to cart, and place orders.
- Track order history and status.

2. ADMINISTRATORS

- Manage product catalog.
- View and process customer orders.
- Monitor inventory and sales reports.

TECHNICAL STACK

- **Frontend:** React.js, Bootstrap, HTML, CSS, JavaScript
- **Backend:** Java 8, Spring Boot, Hibernate, Servlets, JSP
- **Database:** MySQL
- **Server:** Apache Tomcat 9
- **IDE:** Eclipse IDE for Enterprise Java and Web Developers
- **Architecture:** Distributed Microservices with RESTful APIs

SOFTWARE / HARDWARE REQUIREMENT

- **Operating System:** Windows 11
- **Processor:** Intel i3 or higher
- **RAM:** Minimum 8 GB
- **Storage:** 512 GB or more
- **Network:** Stable internet connection for API testing and deployment